Supplementary Material

Growing Clusters Lower the Bar: Profile Accretion in LLM-Based Incremental Entity Resolution

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Code, inputs and every model output behind these tables are available at `https://github.com/xuemian168/profile-accretion`. Percentages refer to hard negatives (negatives A and B pooled) unless stated otherwise; k is the number of verified members shown.

S1 Main FOPPA Design: Uncertainty and Mitigations

Table S1 gives the entity-level uncertainty for the main design and the single-member baselines; Table S2 gives the mitigation results plotted in Fig. 2a of the paper.

Table S1: Uncertainty and baselines. ΔFMR (points) and ΔAUC from $k = 1$ to $k = 16$ with 95% bootstrap intervals over entities; entities whose FMR rises / falls, with the exact sign-test p; FMR (%) of the founding member and mean of the other 15 members judged alone; Spearman ρ between per-entity ΔFMR and number of agents.

Model	ΔFMR [95% CI]	ΔAUC [95% CI]	entities ↑ / ↓	sign-test p	single member: founding / others	Spearman ρ
Qwen2.5-7B	18.2 [14.1, 22.4]	0.05 [0.02, 0.08]	83/6	2×10^{-18}	6.8 / 8.0	−0.02
Qwen2.5-14B	30.5 [25.9, 34.8]	0.08 [0.05, 0.10]	119/0	3×10^{-36}	7.1 / 7.3	+0.03
Qwen2.5-32B	32.3 [27.8, 36.8]	0.08 [0.05, 0.10]	126/1	2×10^{-36}	4.5 / 5.8	+0.13
Qwen3-32B	18.4 [14.1, 22.6]	0.08 [0.05, 0.11]	82/8	1×10^{-16}	15.4 / 16.1	+0.12
OLMo-2-32B	13.5 [10.3, 16.9]	0.05 [0.02, 0.07]	59/2	2×10^{-15}	4.1 / 4.9	+0.11
Mistral-24B	23.7 [19.5, 27.8]	0.08 [0.05, 0.11]	97/0	1×10^{-29}	0.4 / 0.9	+0.07

Table S2: Mitigations at $k = 16$: false-match rate / recall (%) on the controlled FOPPA set. Skeptical: instruction to look for evidence of a different legal entity; last 4: the four most recent members; found.+3: founding member and the last three.

Model	$k = 1$	$k = 16$	skeptical	last 4	found.+3
Qwen2.5-7B	6.8 / 22.6	25.0 / 58.6	16.4 / 45.1	18.0 / 44.4	15.4 / 40.2
Qwen2.5-14B	7.1 / 22.2	37.6 / 86.1	26.7 / 74.1	33.8 / 75.2	32.1 / 78.2
Qwen2.5-32B	4.5 / 18.4	36.8 / 87.6	29.9 / 81.2	29.9 / 71.1	27.4 / 71.8
Qwen3-32B	15.4 / 42.5	33.8 / 81.2	40.4 / 88.7	29.7 / 74.8	28.9 / 76.7
OLMo-2-32B	4.1 / 14.3	17.7 / 44.4	24.4 / 63.9	13.0 / 38.3	13.0 / 33.8
Mistral-24B	0.4 / 2.6	24.1 / 62.4	13.2 / 45.1	10.5 / 33.8	9.8 / 36.8

S2 Prompts

The main experiment uses prompt V0. The user turn lists the members in the order founding member first; Qwen3-32B is run with thinking disabled. P(YES) is the next-token probability mass on YES tokens (any case) divided by the mass on YES and NO tokens.

V0 (main), system.

```
You are an entity-resolution expert for French public-procurement supplier records.  Each
record is 'NAME | CITY POSTCODE'. Two records refer to the same legal entity if they share
the same SIREN (different establishments/branches of one company count as the SAME).
Distinct legal entities in the same corporate group (subsidiaries, sister companies,
franchisees) count as DIFFERENT.
```

V0, user.

```
Known entity - the following {k} record(s) were verified to belong to ONE legal entity:
1.  {member_1}
...
New record:  {query}
Does the new record belong to that same legal entity?  Answer with exactly one word:  YES or
NO.
```

V1 (reworded), system.

```
You work as a data-matching analyst reconciling French public-procurement supplier listings.
Records are formatted as 'NAME | CITY POSTCODE'. Two records denote the same legal entity
exactly when they share an identical SIREN number; subsidiaries, sister firms, and
franchisees that belong to the same corporate group are DIFFERENT legal entities even when
the brand name matches.
```

V1, user.

```
A cluster of {k} confirmed record(s), all previously established as describing one single
legal entity, is listed below:
1.  {member_1}
...
Candidate record under review:  {query}
Is the candidate record describing the same legal entity as the cluster above?  Reply with a
single word, YES or NO.
```

V2 (table, "Answer:" readout), system.

```
You are an entity-resolution expert for French public-procurement supplier records.  Each
record is 'NAME | CITY POSTCODE'. Two records refer to the same legal entity if they share
the same SIREN (different establishments/branches of one company count as the SAME).
Distinct legal entities in the same corporate group (subsidiaries, sister companies,
franchisees) count as DIFFERENT. Format your answer exactly as ''Answer:  YES'' or ''Answer:
NO'', with no other text.
```

V2, user.

```
Verified profile ({k} record(s) confirmed as one legal entity):
| # | Record |
|--|--|
| 1 | {member_1} |
...
Record to classify:  {query}
Does the record to classify belong to the same legal entity as the verified profile above?
```
[assistant turn prefilled with] `Answer:`

V3 (no entity definition), system.

```
You are a helpful assistant.
```

V3, user.

```
These records are known to be about the same company:
- {member_1}
- ...
Here is another record:  {query}
Do these records refer to the same company?  Answer with exactly one word:  YES or NO.
```

The variants were scored at $k \in \{1, 4, 16\}$ for all three queries, plus the 16 single-member judgments of each negative needed for the OR null (10,906 prompts per model and variant); the reversed-order control was not repeated. For V2 the readout is the token after the literal string "Answer:" appended to the chat template. In all variants the YES and NO masses sum the first tokens of YES, Yes and yes (and of NO, No and no), each with and without a leading space. The minimum YES/NO mass per model and variant is reported in Table S3; for the main prompt V0 it is 0.902.

Table S3: Prompt variants on FOPPA (%, except AUC). ΔFMR from $k = 1$ to $k = 16$ with 95% entity-bootstrap interval; mass: minimum YES/NO probability mass over all prompts.

Model	Var.	FMR k=1	FMR k=16	ΔFMR [95% CI]	Rec. k=1	Rec. k=16	AUC k=1	AUC k=16	mass
Qwen2.5-7B	V1	0.0	2.3	+2.3 [+0.9, +3.9]	0.0	5.6	0.68	0.75	1.000
Qwen2.5-7B	V2	3.9	14.1	+10.1 [+6.6, +13.9]	14.7	31.6	0.69	0.72	1.000
Qwen2.5-7B	V3	20.9	43.2	+22.4 [+17.9, +27.1]	36.8	82.3	0.66	0.74	1.000
Qwen2.5-14B	V1	0.0	12.0	+12.1 [+9.0, +15.2]	0.0	41.0	0.63	0.77	0.999
Qwen2.5-14B	V2	3.8	13.3	+9.6 [+7.0, +12.6]	16.2	41.7	0.71	0.77	0.999
Qwen2.5-14B	V3	6.4	38.5	+32.1 [+27.8, +36.8]	13.9	86.1	0.67	0.78	1.000
Qwen2.5-32B	V1	0.0	22.2	+22.2 [+18.2, +26.3]	0.0	71.8	0.71	0.80	1.000
Qwen2.5-32B	V2	5.5	28.6	+23.1 [+19.2, +27.4]	15.8	78.9	0.71	0.79	1.000
Qwen2.5-32B	V3	15.4	39.5	+24.1 [+19.9, +28.4]	36.1	88.0	0.70	0.79	0.996
Qwen3-32B	V1	0.8	26.5	+25.7 [+21.4, +30.1]	1.1	74.1	0.66	0.79	0.992
Qwen3-32B	V2	5.8	27.1	+21.2 [+17.3, +25.4]	20.3	72.6	0.70	0.79	0.999
Qwen3-32B	V3	20.5	41.7	+21.3 [+17.3, +25.4]	47.7	88.0	0.69	0.78	0.995
OLMo-2-32B	V1	0.0	0.4	+0.4 [+0.0, +0.9]	0.4	0.8	0.70	0.73	0.998
OLMo-2-32B	V2	0.2	10.2	+10.0 [+7.1, +13.2]	0.8	28.9	0.71	0.75	0.998
OLMo-2-32B	V3	12.6	30.6	+18.0 [+13.9, +22.4]	32.0	65.0	0.69	0.73	0.998
Mistral-24B	V1	0.0	0.4	+0.4 [+0.0, +0.9]	0.0	0.4	0.68	0.71	0.886
Mistral-24B	V2	14.3	35.5	+21.2 [+16.9, +25.4]	39.1	77.8	0.69	0.75	0.987
Mistral-24B	V3	20.5	47.4	+26.9 [+22.4, +31.6]	43.2	89.5	0.68	0.77	0.979

S3 Street Address Ablation

Each FOPPA agent's street address (field `address` of `Agents.csv`) is inserted between name and city, so a record reads `NAME | ADDRESS | CITY POSTCODE`; the system prompt names the new field. Entities, members, queries and negatives are identical to the main design.

Table S4: FOPPA with street address (%, except AUC). Distinct: negatives whose name differs from every member ($k = 1$ / $k = 16$); OR: OR null at $k = 16$.

Model	FMR k=1	FMR k=16	ΔFMR [95% CI]	distinct	OR	Rec. k=1	Rec. k=16	AUC k=1	AUC k=16
Qwen2.5-7B	4.9	24.2	+19.4 [+15.2, +23.9]	2.1 / 12.4	27.8	18.4	58.6	0.70	0.75
Qwen2.5-14B	4.1	35.3	+31.2 [+26.7, +35.7]	1.4 / 22.1	24.1	16.5	85.0	0.69	0.78
Qwen2.5-32B	0.9	36.3	+35.3 [+30.6, +40.0]	0.7 / 23.2	9.6	0.4	85.3	0.69	0.78
Qwen3-32B	10.0	32.7	+22.7 [+18.4, +27.1]	4.6 / 18.9	34.6	32.3	81.6	0.71	0.78
OLMo-2-32B	2.8	18.2	+15.4 [+11.8, +19.2]	0.5 / 8.5	17.5	11.3	39.1	0.71	0.75
Mistral-24B	0.9	16.5	+15.6 [+12.0, +19.4]	0.5 / 6.9	6.6	3.0	43.6	0.69	0.76

S4 Closed Models

Claude Sonnet 5 (`claude-sonnet-5`) and Claude Haiku 5.5 (`claude-haiku-5-5`) receive prompt V0 for a random 100 of the 266 entities at $k \in \{1, 4, 16\}$ (900 prompts) and the 106 clean/contaminated snowball pairs at $k = 4$ (212 prompts). The main configuration disables extended thinking for both models, so that each answer is the model's direct reply. Sonnet 5 was also run with its default extended reasoning. Each prompt is sampled three times at the API default temperature (no temperature parameter is set); a match is a majority of YES answers among the valid samples, with ties counted as NO, and a confident false match is a YES in all three samples. This unanimity criterion is not on the same scale as the $P(\text{YES}) > 0.99$ criterion used for the open models, and no AUC is computed for the closed models. With thinking disabled, Sonnet 5 returned a parseable answer for all

but 3 of 3,336 requests and Haiku 5.5 for all; with default reasoning, Sonnet 5 returned three valid samples for 1,104 of the 1,112 prompts, the remainder ending with an empty answer after the reasoning budget was spent. A partial Haiku shard run with thinking enabled was discarded and rerun with thinking disabled. Paired $k = 1$ versus $k = 16$ contrasts use exact McNemar tests on negatives observed at both sizes; entity-bootstrap intervals for ΔFMR are 24.0 [17.5, 31.5] points (Sonnet 5, thinking off), 22.0 [15.5, 29.0] (Haiku 5.5) and 20.5 [14.5, 27.5] (Sonnet 5, default reasoning). For the snowball contrast at $k = 4$, acceptance of the intruder's second record rises from 11.3% to 17.0% (Sonnet 5, thinking off, $p = 0.146$), 7.5% to 14.2% (Haiku 5.5, $p = 0.039$) and 13.2% to 24.5% (Sonnet 5, default reasoning, $p = 0.012$).

Table S5: Closed models on 100 FOPPA entities (%), three samples per prompt. Conf.: share of hard negatives answered YES in all three samples; n: hard negatives / positives.

Model	k	FMR	Conf.	Recall	n
Sonnet 5, thinking off	1	0.5	0.5	11.0	200 / 100
	4	12.5	12.0	38.0	200 / 100
	16	24.5	24.5	70.7	200 / 99
Haiku 5.5, thinking off	1	0.0	0.0	0.0	200 / 100
	4	10.0	6.5	26.0	200 / 100
	16	22.0	16.5	53.0	200 / 100
Sonnet 5, default reasoning	1	4.5	3.5	14.0	200 / 100
	4	16.0	14.5	41.0	200 / 100
	16	25.0	25.0	73.0	200 / 100

S5 Published Matching Formats

BatchER-style [1]: the SIREN task description, two fixed demonstrations (one match, one non-match) from an entity outside the 266, and k numbered questions "Record A: member$_i$ | Record B: query" answered in one completion; the decision is the OR of the k answers; greedy decoding. *ComEM-select* [2]: the model chooses among cluster A (k members of the query's entity), cluster B (k members of a fixed unrelated entity) and "none", scored by next-token probabilities over the three option letters; a negative is falsely assigned if A is the most probable option. *LLM-CER-style* [3]: the k members, the query and two fixed distractor records of another entity are numbered and the model returns a partition as lists of record numbers (greedy decoding); the query matches if it is placed with a majority of the members. Parse failures are excluded and reported. The re-implementations keep each format's interaction structure but not each system's blocking, demonstration selection, pre-filtering or hierarchical merging, so absolute rates are not comparable with the original papers; only the change over k is informative. For Qwen2.5-7B, 37.9% of LLM-CER-style outputs could not be parsed within the generation budget, so its estimates in that format are less precise.

Table S7 separates the two sources of the BatchER-style increase. The question about the founding member is identical at $k = 1$ and $k = 16$; at $k = 16$ it is batched with 15 further questions about the same entity.

S6 Second Dataset: TED Czech and Bulgarian Suppliers

Source and identifiers. We use the TED contract-award-notice CSV extract (DG GROW) for 2014–2017, the award years available in our copy, and exclude France to avoid overlap with FOPPA. For each country we measure how often the winner's national identifier (`WIN_NATIONALID`) is filled in and how often it passes the national checksum (Table S8). Czechia (IČO, eight digits with a mod-11 check digit) and Bulgaria (EIK/Bulstat, nine digits with a mod-11 check digit) combine coverage above a third with checksum validity above 98% and enough identifiers with 17 or more distinct records. Companies House numbers (United Kingdom) and NIP (Poland) are filled in for about 2% of winner rows. *Records and entities.* A record is a distinct (name, street address, postcode, town) tuple shown as `NAME | STREET ADDRESS | POSTCODE TOWN | COUNTRY`; an entity is a checksum-valid, normalised identifier. The 102 identifiers with at least 17 distinct records (16 Czech, 86 Bulgarian) form the entity set; results are pooled because the Czech subset is too small for a separate test. In 22 (negative A) and 6 (negative B) cases the negative's name equals a member's name. Members and the held-out positive are drawn as in FOPPA. Negatives A and B are the most name-similar records of other identifiers among all 17,267 distinct

Table S6: Published matching formats on FOPPA: rate (%) of assignment to the profile at $k = 1 \rightarrow k = 16$ for negatives A and B (false matches) and positives (recall); exact McNemar p for $k = 1$ versus $k = 16$ (negA / negB); parse: share of unparseable outputs (%).

Format	Model	negA	negB	pos	McNemar p	parse
BatchER	Qwen2.5-7B	$22.9 \rightarrow 54.1$	$16.9 \rightarrow 49.2$	$51.5 \rightarrow 83.5$	5×10^{-23} / 5×10^{-23}	0.0
BatchER	Qwen2.5-14B	$24.1 \rightarrow 43.2$	$18.4 \rightarrow 36.5$	$53.4 \rightarrow 78.9$	4×10^{-13} / 1×10^{-10}	0.0
BatchER	Qwen2.5-32B	$23.7 \rightarrow 47.7$	$13.9 \rightarrow 37.6$	$48.5 \rightarrow 87.2$	1×10^{-19} / 2×10^{-19}	0.0
BatchER	Qwen3-32B	$19.5 \rightarrow 41.7$	$12.4 \rightarrow 35.7$	$44.0 \rightarrow 78.9$	5×10^{-17} / 6×10^{-17}	0.0
BatchER	OLMo-2-32B	$23.3 \rightarrow 43.6$	$16.9 \rightarrow 36.1$	$50.4 \rightarrow 73.7$	3×10^{-12} / 2×10^{-11}	0.0
BatchER	Mistral-24B	$45.9 \rightarrow 63.9$	$40.6 \rightarrow 59.4$	$71.1 \rightarrow 91.4$	3×10^{-11} / 3×10^{-11}	0.0
ComEM-select	Qwen2.5-7B	$7.5 \rightarrow 42.5$	$8.6 \rightarrow 35.0$	$19.9 \rightarrow 71.4$	5×10^{-27} / 4×10^{-17}	0.0
ComEM-select	Qwen2.5-14B	$28.6 \rightarrow 50.4$	$21.4 \rightarrow 43.2$	$55.3 \rightarrow 88.0$	8×10^{-16} / 8×10^{-14}	0.0
ComEM-select	Qwen2.5-32B	$28.2 \rightarrow 57.1$	$24.1 \rightarrow 50.8$	$60.5 \rightarrow 91.0$	1×10^{-23} / 2×10^{-19}	0.0
ComEM-select	Qwen3-32B	$21.1 \rightarrow 51.9$	$13.2 \rightarrow 45.9$	$47.0 \rightarrow 88.3$	9×10^{-24} / 1×10^{-26}	0.0
ComEM-select	OLMo-2-32B	$9.0 \rightarrow 53.0$	$5.3 \rightarrow 47.7$	$18.8 \rightarrow 85.0$	1×10^{-35} / 6×10^{-33}	0.0
ComEM-select	Mistral-24B	$1.1 \rightarrow 42.9$	$1.1 \rightarrow 32.3$	$11.3 \rightarrow 78.9$	8×10^{-34} / 2×10^{-25}	0.0
LLM-CER	Qwen2.5-7B	$52.0 \rightarrow 21.6$	$47.8 \rightarrow 21.1$	$74.2 \rightarrow 47.1$	6×10^{-13} / 8×10^{-10}	37.9
LLM-CER	Qwen2.5-14B	$24.4 \rightarrow 12.4$	$16.9 \rightarrow 9.4$	$49.2 \rightarrow 32.8$	9×10^{-6} / 0.002	0.2
LLM-CER	Qwen2.5-32B	$15.3 \rightarrow 14.9$	$11.2 \rightarrow 11.9$	$40.0 \rightarrow 32.3$	0.880 / 1.000	1.5
LLM-CER	Qwen3-32B	$21.1 \rightarrow 7.5$	$15.1 \rightarrow 6.4$	$47.4 \rightarrow 27.4$	1×10^{-6} / 6×10^{-6}	0.1
LLM-CER	OLMo-2-32B	$42.2 \rightarrow 6.0$	$37.8 \rightarrow 3.4$	$71.6 \rightarrow 12.4$	2×10^{-22} / 1×10^{-21}	2.3
LLM-CER	Mistral-24B	$55.3 \rightarrow 17.3$	$53.0 \rightarrow 12.0$	$81.2 \rightarrow 46.2$	9×10^{-24} / 6×10^{-28}	0.0

Table S7: BatchER-style format: share (%) of hard negatives for which the founding-member question is answered YES when asked alone ($k = 1$) and when batched with 15 further members ($k = 16$); paired changes NO→YES / YES→NO with exact McNemar p.

Model	$k = 1$	$k = 16$	changes	p
Qwen2.5-7B	19.9	22.6	26 / 12	0.034
Qwen2.5-14B	21.2	21.2	17 / 17	1.000
Qwen2.5-32B	18.8	26.3	43 / 3	5×10^{-10}
Qwen3-32B	16.0	21.2	35 / 7	2×10^{-5}
OLMo-2-32B	20.1	18.0	12 / 23	0.090
Mistral-24B	43.2	36.3	17 / 54	1×10^{-5}

(name, address, postcode, town) records with a checksum-valid Czech or Bulgarian identifier, using character 3–4-gram TF-IDF over Unicode-normalised names so that Cyrillic names are retained. The system prompt replaces the SIREN definition by the national registration number. Scoring, nested profiles, OR null, mitigations and the snowball contrast follow the main design.

Table S8: National identifiers among TED winner records, 2014–2017: winner rows, share with a non-empty identifier (%), share of non-empty identifiers passing the checksum (%), and identifiers with at least 17 distinct records. Countries without a checkable scheme are omitted.

Country	Identifier	winner rows	filled	valid	entities
CZ	IČO	54,231	35.0	98.5	16
BG	EIK/Bulstat	51,666	34.3	99.2	86
SK	IČO	12,645	38.5	99.5	1
FI	Y-tunnus	14,506	33.8	96.6	2
SE	Organisationsnummer	41,419	19.5	97.6	4
ES	CIF	57,561	16.6	87.7	–
UK	Companies House CRN	148,043	2.2	25.5	–
PL	NIP	437,272	1.5	39.8	–

Table S9: TED Czech and Bulgarian suppliers, 102 entities (%, except AUC). Distinct: negatives whose name differs from every member ($k = 1$ / $k = 16$); OR: OR null at $k = 16$; snowball: acceptance of the intruder's second record at $k = 4$, clean → contaminated.

Model	FMR k=1	FMR k=16	ΔFMR [95% CI]	distinct	OR	Rec. k=1	Rec. k=16	AUC k=1	snowball
Qwen2.5-7B	23.0	13.7	−9.3 [−14.2, −4.4]	11.9 / 3.4	35.3	99.0	92.2	0.93	2.4 → 8.1
Qwen2.5-14B	14.2	14.7	+0.5 [−1.5, +2.9]	2.3 / 4.0	17.6	77.5	92.2	0.91	2.4 → 39.8
Qwen2.5-32B	13.7	15.7	+2.0 [−0.0, +4.4]	1.7 / 4.5	19.1	89.2	94.1	0.92	2.4 → 67.5
Qwen3-32B	14.2	14.2	+0.0 [−2.5, +2.5]	2.3 / 3.4	21.6	96.1	92.2	0.92	2.4 → 61.8
OLMo-2-32B	15.7	14.2	−1.5 [−3.4, +0.5]	4.0 / 4.0	24.5	97.1	89.2	0.92	3.3 → 31.7
Mistral-24B	12.7	14.7	+2.0 [−1.0, +4.9]	2.8 / 4.0	19.6	84.3	95.1	0.93	3.3 → 69.9

S7 Remedies in Online Linking

All remedies run on the 80 confusable blocks of the paper (2,572 records) with the candidate generation described there. Size-calibrated thresholds are fit on the 186 FOPPA entities that do not occur in the blocks: for each profile size $k \in \{1, 2, 4, 8, 16\}$ the smallest threshold is chosen whose pooled hard-negative false-match rate does not exceed τ; a cluster of size s uses the threshold of the largest $k \leq s$. The first fit (*size-calibrated* in Table S11) targets the $k = 1$ false-match rate at threshold 0.5; the sweep in Fig. S1 uses $\tau \in \{0.05, 0.1, 0.15, 0.2, 0.3\}$. The audit re-scores every non-founding member against the founding member alone when a cluster reaches 4, 8 or 16 members and moves it to a singleton if $P(\text{YES}) < 0.5$; the soft audit requires in addition that the member's leave-one-out profile score is below 0.5 and uses a founder threshold of 0.05. Intervals are block bootstrap (2,000 resamples), paired where two representations are compared. Matched-recall comparisons interpolate linearly between swept points and are reported only where both curves cover the target recall (0.2, 0.3, 0.4).

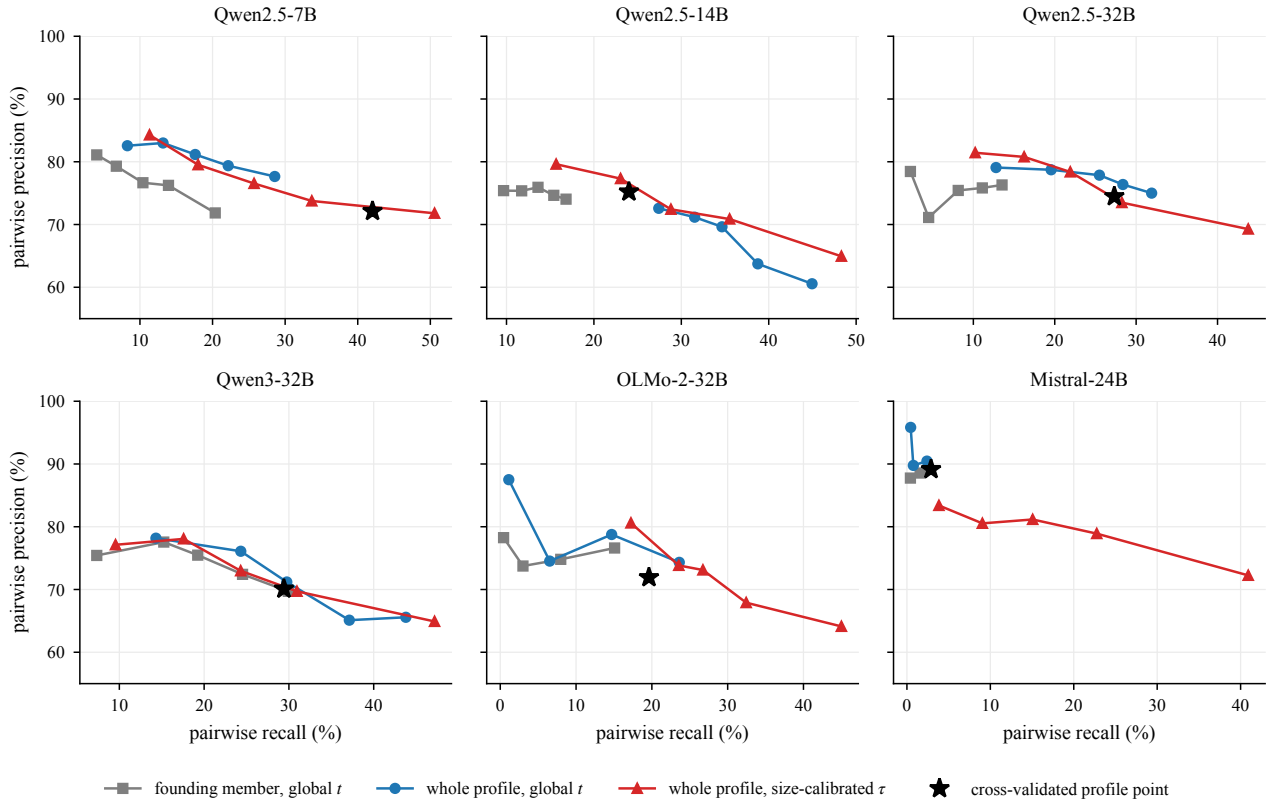


Figure S1: Pairwise precision and recall of online linking for the founding-member and whole-profile representations under global thresholds $t \in \{0.5, 0.8, 0.95, 0.99, 0.999\}$ and size-calibrated thresholds $\tau \in \{0.05, \ldots, 0.3\}$. Stars: cross-validated profile operating point (Table S10). Points with recall below 0.2% are omitted.

Table S10: Precision-matched global threshold for the whole profile (the remedy recommended in the paper), five-fold cross-validation over blocks (20 repetitions). Precision / recall (%) of the founding-member rule at $t = 0.5$ and of the selected whole-profile point on held-out blocks; differences in points with 95% block-bootstrap intervals; recall ratio profile / founding member.

Model	founding P / R	profile P / R	ΔP [95% CI]	R ratio	ΔR [95% CI]
Qwen2.5-7B	71.8 / 20.4	72.1 / 42.0	+0.3 [−8.1, +7.1]	2.06	+21.7 [+16.7, +26.5]
Qwen2.5-14B	74.0 / 16.8	75.2 / 24.0	+1.2 [−3.5, +5.8]	1.43	+7.2 [+4.4, +10.1]
Qwen2.5-32B	76.3 / 13.5	74.5 / 27.3	−1.8 [−6.2, +5.6]	2.02	+13.8 [+7.8, +16.9]
Qwen3-32B	69.7 / 29.9	70.1 / 29.4	+0.4 [−4.6, +4.7]	0.98	−0.5 [−3.1, +3.1]
OLMo-2-32B	76.6 / 15.1	71.9 / 19.6	−4.7 [−11.4, +3.1]	1.30	+4.5 [+2.0, +6.4]
Mistral-24B	88.6 / 1.6	89.1 / 2.9	+0.6 [−1.9, +7.4]	1.83	+1.3 [+0.1, +1.8]

Table S11: Online linking at the default operating points (%). Size-calibrated: per-size thresholds targeting the $k = 1$ false-match rate (not the recommended remedy; the precision-matched global threshold is in Table S10). Mixed: share of output clusters containing records of more than one SIREN; calls: LLM calls per record, including audit calls.

Model	representation	precision	recall	F_1	mixed	calls
Qwen2.5-7B	founding member	71.8	20.4	31.7	7.2	2.40
	whole profile	77.7	28.6	41.8	7.8	2.39
	size-calibrated	80.8	14.8	25.0	6.9	2.44
	profile + audit	76.8	22.7	35.1	7.6	3.03
	calibrated + audit	80.0	14.1	23.9	7.2	2.79
	soft audit	77.4	27.9	41.0	7.7	3.42
Qwen2.5-14B	founding member	74.0	16.8	27.4	6.6	2.43
	whole profile	60.6	45.0	51.6	6.8	2.26
	size-calibrated	78.8	19.0	30.6	5.0	2.43
	profile + audit	68.7	31.5	43.2	8.4	3.27
	calibrated + audit	78.8	16.4	27.1	5.3	2.80
	soft audit	60.2	43.5	50.5	6.6	3.50
Qwen2.5-32B	founding member	76.3	13.5	23.0	5.3	2.45
	whole profile	75.0	31.9	44.8	5.3	2.41
	size-calibrated	81.2	10.0	17.9	4.4	2.47
	profile + audit	75.3	22.5	34.7	5.4	3.05
	calibrated + audit	78.4	8.0	14.5	4.4	2.63
	soft audit	75.4	31.8	44.7	5.3	3.42
Qwen3-32B	founding member	69.7	29.9	41.9	11.7	2.32
	whole profile	65.6	43.8	52.5	10.0	2.24
	size-calibrated	72.9	24.8	37.0	9.2	2.40
	profile + audit	66.6	36.6	47.2	10.2	3.14
	calibrated + audit	73.3	23.1	35.1	9.1	2.89
	soft audit	65.5	43.4	52.2	10.0	3.59
OLMo-2-32B	founding member	76.6	15.1	25.2	5.1	2.45
	whole profile	74.3	23.6	35.8	5.4	2.43
	size-calibrated	80.5	16.4	27.3	5.3	2.45
	profile + audit	71.6	20.5	31.9	5.4	2.95
	calibrated + audit	80.4	15.6	26.1	5.3	2.78
	soft audit	74.3	23.5	35.7	5.4	3.19
Mistral-24B	founding member	88.6	1.6	3.1	1.0	2.50
	whole profile	90.5	2.4	4.7	1.0	2.50
	size-calibrated	88.7	0.9	1.8	0.9	2.50
	profile + audit	90.7	2.3	4.4	1.0	2.54
	calibrated + audit	88.7	0.9	1.8	0.9	2.51
	soft audit	90.7	2.5	4.8	1.0	2.58

S8 Additional Checks

Table S12 reports the main FOPPA design separately for the two negatives. Negative A is chosen for its similarity to the founding member, which is present in every profile; negative B is chosen for its similarity to one of members 9–16, which first enter the profile at $k = 16$. Table S13 restricts FOPPA and TED to negatives whose canonical-name cosine to the member they were chosen for is at least 0.7 (median cosine of negative A: FOPPA 0.76, TED 0.60). Table S14 repeats the online-linking comparison with BCubed precision, recall and F_1 [4], averaged

over records; the precision-matched profile point is selected on pairwise precision exactly as in Table S10 and evaluated with BCubed on the held-out blocks. As a non-LLM reference, the character 3–4-gram TF-IDF cosine between displayed names (vectoriser fit on all displayed records of the study) separates held-out members from hard negatives with an AUC of 0.69 against the founding member and 0.75 as the maximum over the 16 members; on full displayed records the AUCs are lower. In online linking, 90% of Qwen2.5-32B's decisions in clusters of eight or more members and 85% of Qwen2.5-14B's have $P(\text{YES}) \geq 0.9999$.

Table S12: Main FOPPA design by negative type: false-match rate (%) at $k = 1 \rightarrow k = 16$ and paired changes NO→YES / YES→NO (266 negatives each).

Model	negative A	changes	negative B	changes
Qwen2.5-7B	$9.4 \rightarrow 27.8$	52 / 3	$4.1 \rightarrow 22.2$	52 / 4
Qwen2.5-14B	$9.4 \rightarrow 42.1$	87 / 0	$4.9 \rightarrow 33.1$	75 / 0
Qwen2.5-32B	$4.9 \rightarrow 42.1$	100 / 1	$4.1 \rightarrow 31.6$	73 / 0
Qwen3-32B	$16.2 \rightarrow 37.2$	64 / 8	$14.7 \rightarrow 30.5$	45 / 3
OLMo-2-32B	$5.3 \rightarrow 19.2$	39 / 2	$3.0 \rightarrow 16.2$	36 / 1
Mistral-24B	$0.0 \rightarrow 26.7$	71 / 0	$0.8 \rightarrow 21.4$	55 / 0

Table S13: Negatives with name cosine at least 0.7 to the member they were chosen for: false-match rate (%) at $k = 1 \rightarrow k = 16$, paired changes and exact McNemar p.

Data	Model	n	FMR	changes	p
FOPPA	Qwen2.5-7B	331	$9.7 \rightarrow 29.0$	69 / 5	2×10^{-15}
	Qwen2.5-14B	331	$9.1 \rightarrow 45.6$	121 / 0	8×10^{-37}
	Qwen2.5-32B	331	$6.3 \rightarrow 43.5$	124 / 1	6×10^{-36}
	Qwen3-32B	331	$19.3 \rightarrow 40.8$	79 / 8	8×10^{-16}
	OLMo-2-32B	331	$5.7 \rightarrow 22.1$	57 / 3	6×10^{-14}
	Mistral-24B	331	$0.0 \rightarrow 28.4$	94 / 0	1×10^{-28}
TED	Qwen2.5-7B	60	$56.7 \rightarrow 43.3$	1 / 9	0.021
	Qwen2.5-14B	60	$46.7 \rightarrow 48.3$	3 / 2	1.000
	Qwen2.5-32B	60	$45.0 \rightarrow 51.7$	5 / 1	0.219
	Qwen3-32B	60	$46.7 \rightarrow 46.7$	3 / 3	1.000
	OLMo-2-32B	60	$48.3 \rightarrow 45.0$	1 / 3	0.625
	Mistral-24B	60	$41.7 \rightarrow 46.7$	5 / 2	0.453

Length control. Each query is scored against the founding member alone, against the founding member repeated 4 and 16 times, and against the 16 distinct members, in one run per model with the main prompt V0 (3,192 prompts per model). The repeated profiles are as long as profiles of 4 and 16 members but carry no additional content. The single-member and 16-member rates differ slightly from Table I of the paper because the prompts were batched differently.

Readout precision. $P(\text{YES})$ is computed in 32-bit floating point and rounds to exactly 1 once the YES–NO log-odds exceed about 17. The four Qwen models were rescored with the log-odds $\log \sum_{\text{YES}} e^z - \log \sum_{\text{NO}} e^z$ computed from the logits z (Table S18). Decisions at 0.5 agree with the original run for at least 99.9% of prompts. For Qwen2.5-32B the 190 profile scores at $P(\text{YES}) = 1$ spread over log-odds of 17–25 (46 distinct values at three decimals, since the logits are computed in bfloat16), so a matched-recall threshold exists in every fold.

Other checks. Table S16 gives the LLM calls per record of the three online-linking representations; pairwise max scores every shown member of each candidate cluster. Table S17 splits the TED result by country. Confident false matches show the same ordering at cut-offs of 0.95 and 0.999 as at 0.99 (file `data/derived/review/conf_thresholds.csv` in the repository). The LLM-CER-style parse failures of Qwen2.5-7B are concentrated at $k = 4$ (81–89% of outputs, against 4–7% at $k = 1$ and 22–23% at $k = 16$), where the model wrote long explanations before the partition. A Holm correction over the 56 model-level McNemar tests behind Table III of the paper leaves every count of significant increases unchanged (file `data/derived/review/holm_table3.csv`).

Table S14: Online linking with BCubed precision, recall and F_1 (%).

Model	representation	precision	recall	F_1
Qwen2.5-7B	founding member	88.8	33.7	48.9
	pairwise max	86.8	40.9	55.6
	whole profile	88.5	39.5	54.6
	precision-matched profile	83.0	50.3	62.6
Qwen2.5-14B	founding member	90.0	31.3	46.5
	pairwise max	87.8	38.7	53.7
	whole profile	82.1	53.6	64.9
	precision-matched profile	90.4	36.3	51.7
Qwen2.5-32B	founding member	92.5	28.7	43.8
	pairwise max	90.9	35.0	50.6
	whole profile	89.6	42.2	57.4
	precision-matched profile	90.7	38.8	54.4
Qwen3-32B	founding member	84.5	42.2	56.3
	pairwise max	81.2	55.8	66.2
	whole profile	82.0	53.1	64.4
	precision-matched profile	86.2	40.9	55.5
OLMo-2-32B	founding member	92.3	29.1	44.2
	pairwise max	91.5	31.3	46.6
	whole profile	90.5	35.3	50.8
	precision-matched profile	90.7	33.1	48.5
Mistral-24B	founding member	98.9	18.8	31.6
	pairwise max	98.8	19.1	32.1
	whole profile	98.8	19.5	32.6
	precision-matched profile	98.7	19.6	32.7

Table S15: Length control on FOPPA: false-match rate (%) on 532 hard negatives with the founding member alone, repeated 4 and 16 times, and with 16 distinct members; paired changes NO→YES / YES→NO and exact McNemar p for one versus 16 copies; recall (%) for one member / 16 copies / 16 members.

Model	$k = 1$	4×	16×	16 members	changes (16×)	p	recall
Qwen2.5-7B	7.1	0.2	0.2	24.8	1 / 38	1×10^{-10}	22.9 / 0.4 / 58.6
Qwen2.5-14B	7.1	3.2	2.6	37.8	4 / 28	2×10^{-5}	21.1 / 12.0 / 86.1
Qwen2.5-32B	4.5	1.5	1.7	37.2	4 / 19	0.003	18.8 / 3.4 / 88.0
Qwen3-32B	15.2	4.7	3.8	33.8	0 / 61	9×10^{-19}	42.5 / 12.8 / 81.2
OLMo-2-32B	4.1	0.6	0.6	17.7	0 / 19	4×10^{-6}	14.3 / 1.1 / 45.1
Mistral-24B	0.4	0.6	1.5	23.7	6 / 0	0.031	3.0 / 6.4 / 62.8

Table S16: LLM calls per record in online linking (80 blocks, 2,572 records).

Model	founding member	pairwise max	whole profile
Qwen2.5-7B	2.40	7.09	2.39
Qwen2.5-14B	2.43	7.09	2.26
Qwen2.5-32B	2.45	6.01	2.41
Qwen3-32B	2.32	9.43	2.24
OLMo-2-32B	2.45	5.60	2.43
Mistral-24B	2.50	2.82	2.50

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Table S17: TED by country: false-match rate (%) at $k = 1 \rightarrow k = 16$ and paired changes NO→YES / YES→NO (Czechia 32, Bulgaria 172 hard negatives).

Model	CZ	changes	BG	changes
Qwen2.5-7B	$18.8 \rightarrow 12.5$	0 / 2	$23.8 \rightarrow 14.0$	1 / 18
Qwen2.5-14B	$9.4 \rightarrow 9.4$	0 / 0	$15.1 \rightarrow 15.7$	3 / 2
Qwen2.5-32B	$9.4 \rightarrow 15.6$	2 / 0	$14.5 \rightarrow 15.7$	3 / 1
Qwen3-32B	$9.4 \rightarrow 9.4$	0 / 0	$15.1 \rightarrow 15.1$	3 / 3
OLMo-2-32B	$15.6 \rightarrow 15.6$	1 / 1	$15.7 \rightarrow 14.0$	0 / 3
Mistral-24B	$6.2 \rightarrow 15.6$	3 / 0	$14.0 \rightarrow 14.5$	3 / 2

Table S18: Log-odds rescoring of the main FOPPA design: agreement of decisions with the original run (%), number of $k = 16$ profile scores with $P(\text{YES}) = 1$ and their distinct log-odds values, AUC at $k = 16$ from $P(\text{YES})$ / log-odds, and matched recall (%) at $k = 1$ / $k = 16$ on the log-odds scale.

Model	agree	$P = 1$	distinct	AUC k=16	matched recall
Qwen2.5-7B	99.9	0	0	0.749 / 0.749	22.6 / 15.7
Qwen2.5-14B	99.9	32	12	0.780 / 0.780	21.4 / 25.0
Qwen2.5-32B	99.9	190	46	0.800 / 0.797	18.0 / 10.7
Qwen3-32B	99.9	0	0	0.801 / 0.801	42.5 / 51.9

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